

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Medium

Question

The speed of sound in dry air, v , can be modeled by the formula $v = 331.3 + 0.606T$, where T is the temperature in degrees Celsius and v is measured in meters per second. Which of the following correctly expresses T in terms of v ?

Answer

- A. $T = \frac{v + 0.606}{331.3}$
- B. $T = \frac{v - 0.606}{331.3}$
- C. $T = \frac{v + 331.3}{0.606}$
- D. $T = \frac{v - 331.3}{0.606}$

Correct Answer: D

Rationale

Choice D is correct. To express T in terms of v , subtract 331.3 from both sides of the equation, which gives $v - 331.3 = 0.606T$. Dividing both sides of the equation by 0.606 gives $\frac{v - 331.3}{0.606} = T$.

Choices A, B, and C are incorrect and are the result of incorrect steps while solving for T .